SUBSTRATE SCANNING ENGINE: GEOMETRIC PARAMETER GENERATION FOR L-FUNCTIONS VIA ELEMENTARY VISUALIZATION AND SYSTEMATIC ELIMINATION

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Abstract. We describe a modular computational engine that takes any Euler product and produces geometric parameters for machine learning without statistical assumptions. The engine maps the Euler product to a toroidal embedding, decomposes the resulting dynamics into three separable scales (drift, spiral, continued-fraction modulation), constructs the electrostatic potential $V(\sigma, t) = -\log|-\zeta'/\zeta|$, computes its Laplacian (confirming harmonicity between zeros at 0.07% median residual), and extracts 29 deterministic features per query point. Applied to the Riemann zeta function, the engine discriminates zeros from non-zeros with 5.6σ separation on phase coherence alone, using only the partial Euler product.

The central decomposition is $\theta^*(k, \gamma) = f(b) \cdot \cos(\ln(p) \cdot \gamma)$, where $f(b) = b/(b+1)$ is the natural per-step contraction (universal, amplitude-only) and $\cos(\ln(p) \cdot \gamma)$ is the phase modulation (specific to each zero). Three independent tests confirm that the arithmetic channel (factorization of $p - 1$) is orthogonal to both amplitude and phase channels, establishing that shared algebra does not imply shared information.

The engine is demonstrated on 16 domains: the Riemann zeta function, Dirichlet L-functions, elliptic curve L-functions, Cayley graph spectra, 4-port differential S-parameters (signal integrity), vibration spectra (rotating machinery fault diagnosis), 2D physical cavities (antenna/acoustic/laser optimization), visible-light spectroscopy (colorimetry), general graph analysis (social networks, supply chains, bot detection), abstract syntax trees (code clone detection, refactoring), phonon spectra (materials design, thermoelectric screening), 3D wavelet subband analysis (volumetric data), turbulence cascade classification, 2D image spectral analysis (forgery detection), spectral pathfinding (graph navigation), and elliptic curve cryptographic auditing—with zero architectural changes between domains. The constraint is precise: the system must have a linear operator with discrete spectrum. Under this condition, the motor generates, ML selects, and domain physics verifies. Estimated computational savings range from 10× (BSD verification) to 10,000× (materials design: 29,241 compositions screened in 32 seconds vs. DFT).

No result in this paper is claimed as mathematically novel. The contribution is methodological: a systematic pipeline for geometric exploration that produces falsifiable outputs, documents its own overclaims, and eliminates impossible approaches at visual speed.

0. Errata and Methodological Transparency

This section appears first because all subsequent claims should be read in light of these corrections. During the development of this engine, seven overclaims were produced and corrected within the same working sessions. Each is documented below with the quantitative evidence that falsified it.

0.1 Overclaims Corrected.

(a) **"Exactly as $f(b)$"** — the per-step contraction $f(b) = b/(b+1)$ is algebraically exact, but the cumulative exponent is -1.4 ± 0.1, not

-1.0. The discrepancy arises because per-step exactness does not imply exact geometric series accumulation.

(b) **Spiral exponent** $\alpha = -0.96$ — artifact of fixed-reference measurement ($P(N_{200})$ as anchor). Reference-independent methods (moving center, consecutive differences) give $\alpha = -1.4 \pm 0.1$ consistently across 5 methods and 8 zeros.

(c) **Poincaré compactification produces convergence to** ∂B^3 — incorrect. Since $|P(N)|$ decreases with N (the normalizing sum $Z(N)$ grows in the denominator), the compactified points move toward the origin of B^3, not toward the boundary. The gap $1 - |\tilde{U}_P|$ grows from 0.59 ($N = 10$) to 0.70 ($N = 2000$). The algebraic identity $f(b) = \text{Poincaré}_{1D}$ remains valid; the geometric application to these trajectories was the overclaim.

(d) **The** $f(b)$ **bridge between digit-sum and toroidal channels** — empirically falsified by the dual test ($v_2(p-1)$ as cross-channel predictor, 5132 primes): $v_2(p-1)$ predicts digit-sum deficit with $\rho = -0.43$ but has $\rho < 0.01$ for all toroidal quantities.

(e) $\gamma \cdot |\omega| \approx$ **constant** — CV = 36%, not constant. Removed.

(f) **The cone of radial directions in** B^3 **closes with** N — it does not. Semi-angle decays as $N^{-0.19}$, too slow for convergence.

(g) **"The zeros exist as points on** ∂B^3**"** — reification of the embedding. The zeros exist as zeros of the analytic continuation of $\zeta(s)$. The compactified trajectories point toward boundary positions that encode the zeros; they do not constitute the zeros.

The ratio of eliminated claims to surviving results is 7:1. This ratio is characteristic of honest computational phenomenology and is the primary diagnostic of the methodology.

1. Introduction: t as Parameter

The factorization

$$p^{-s} = p^{-\sigma} \cdot e^{-i \ln(p) \cdot t} \tag{1}$$

is a fact, not a choice. The parameter t traces the critical line; each prime generates a mode with frequency $\omega = \ln(p)$; the zeros of the Riemann zeta function are the values of t where all phases conspire to annihilate the sum. Everything else in this paper—the embedding, the three scales, the potential landscape, the envelope decomposition—is scaffolding around this single equation.

The Euler product $\prod_p (1 - p^{-s})^{-1}$ converges absolutely for $\sigma > 1$. At $\sigma = 1/2$, the partial products do not converge (Conrad [2], extending earlier work of Goldfeld and Kurokawa et al.). The non-convergence is not a technical obstacle—it is the fundamental structural fact. Kurokawa's "Deep Riemann Hypothesis" asserts that convergence of partial Euler products at

the center of the critical strip would imply something strictly deeper than the generalized Riemann hypothesis.

This paper describes a computational engine that takes any Euler product and produces its geometric atlas: a collection of deterministic features derived from the product's structure, suitable as input for machine learning without statistical assumptions.

2. The Geometric Engine

The engine consists of eight modules, each independently testable. The input is an Euler product (abstract base class); the output is a 29-dimensional feature vector per query point (σ, t, K).

2.1. **Module 1: Euler Product.** Abstract interface with methods: `primes(K)`, `local_factor(p,s)`, `log_deriv_term(p,s)`, `arithmetic_channel(p)`. Concrete implementations provided for $\zeta(s)$, $L(s,\chi)$, and $L(s,E)$. The `arithmetic_channel` method returns features specific to each L-function: $v_2(p-1)$ and $\mathrm{ord}_2(p)/(p-1)$ for ζ; $\chi(p)$ for Dirichlet; a_p for elliptic curves.

2.2. **Module 2: Toroidal Embedding.** Maps the Euler product to weighted coordinates on the mixed torus $\mathbb{T}^2_{(p_k,p_{k+1})}$ via $p^{-s} = p^{-\sigma} \cdot e^{-i\omega t}$. The embedding is standard (Bohr torus, 1920s). What is specific to this work is the pair-weighted parametrization and the decomposition into three scales.

2.3. **Module 3: Three Scales.** The trajectory $P(N)$ decomposes into: Scale 1 (drift, $\sim 1/\ln N$, divergent by Mertens), Scale 2 (spiral, $\sim N^{-1.4}$, reference-independent), Scale 3 (CF modulation, classical Khinchin). The decomposition is specific to this embedding; the scales may not have counterparts in other parametrizations.

2.4. **Module 4: Envelope Decomposition.** The natural per-step weight is $f(b_k) = b_k/(1+b_k)$ where $b_k = p_k^{-\sigma}/Z_k$. Optimization of weighted log-derivative loss over known zeros produces optimal weights $\theta^*(k,\gamma)$. The central result of the pipeline:

$$\theta^*(k,\gamma) = f(b_k) \cdot \big[A + B\cos(\ln(p_k)\cdot\gamma) + C\sin(\ln(p_k)\cdot\gamma)\big] \tag{2}$$

where $f(b)$ is universal (amplitude decay) and the trigonometric factor is zero-specific (phase tuning). R^2 ranges from 0.09 ($K = 20$) to 0.31 ($K = 30$, γ_3). The phase factor is the numerical fingerprint of Connes' prolate operator.

2.5. **Module 5: Potential Landscape.** $V(\sigma,t) = -\log|-\zeta'/\zeta(\sigma+it)|$ is the electrostatic potential with charges at the zeros. Between zeros, $\Delta V \approx 0$ (median $|V|/|V| = 0.0007$ over 458 grid points). Near zeros, ΔV diverges. The integral $\oint V\,dA$ around each zero approaches 2π in principle (Dirac delta charges), though for the partial product the charges are diffuse rather than point-like.

2.6. **Module 6: Weil Spectrum.** Discretized Weil quadratic form $W_{jk} = \sqrt{\ln p_j} \cdot \sqrt{\ln p_k} \cdot p_j^{-1/4} \cdot p_k^{-1/4} \cdot \cos\big((\ln p_j - \ln p_k) \cdot \gamma\big)$. Eigenvalues computed by Jacobi iteration for $K \leq 150$. The minimum eigenvalue detects the zero; the spectral gap measures resolution.

2.7. **Module 7: Channel Orthogonality.** Tests independence between Canal A (arithmetic: v_2, ord_2, χ, a_p), Canal B (amplitude: $\ln p$, $p^{-1/2}$), and Canal C (phase: $\cos(\ln p \cdot \gamma)$, $\sin(\ln p \cdot \gamma)$). Maximum Spearman ρ between column pairs. For ζ: $A \perp B$ ($\rho < 0.28$), $A \perp C$ ($\rho < 0.33$), B not orthogonal to C (share $\ln p$).

2.8. **Module 8: Feature Vector.** Combines all modules into a 29-dimensional vector: 3 meta (σ, t, K), 4 torus, 6 potential (V, ∇V, ΔV, harmonicity), 5 Weil (eigenvalues), 6 envelope ($f(b)$ stats, phase coherence), 2 Poincaré (radius, gap), 3 channel (ρ cross-correlations). Output is a `numpy` array, deterministic, reproducible, and computable in $\sim$8 ms per query.

3. Results

3.1. **Zero Discrimination.** At $K = 30$, $\sigma = 0.5$, the three features with highest zero/non-zero separation are: phase coherence (5.6σ), V potential (2.4σ), and gradient magnitude (0.75σ). Phase coherence alone classifies zeros with zero training. This is the Canal C signal: at a zero, the phases $\ln(p) \cdot \gamma$ conspire, producing high coherence; away from zeros, they cancel.

3.2. **Envelope Factorization.** Per-zero optimal weights are ZERO-SPECIFIC (mean pairwise $|r| = 0.14$). The third axis beyond Canal A and B is $\cos(\ln(p) \cdot \gamma)$—the phase of the target zero evaluated at each prime ($\rho = -0.59$, $p = 6.6 \times 10^{-4}$ at $K = 30$). Canal A (v_2, ord_2) is absent from the optimal weights at every K tested ($r < 0.1$). The optimal operator for zero detection is $f(b) \cdot \cos(\omega)$, which is purely spectral.

3.3. **The σ Transition.** Animating σ from 1.2 to 0.3 shows the convergence/divergence transition as a continuous geometric deformation: the spiral passes from closed (convergent product) to open (divergent drift). The transition point is the boundary of the Euler product's convergence region, visible as the moment the spiral stops closing. This is the prime number theorem made curve: the spiral does not close at $\sigma = 1/2$ because $\sum 1/p$ diverges (Mertens), and the pseudo-randomness of primes decreases with p, visible as increasing coherence of consecutive steps.

3.4. **Geodesics.** Gradient descent on $V(\sigma, t)$ from any $\sigma > 1/2$ converges to $\sigma = 1/2$. The geodesics channel between zero-wells, attracted by the saddle points. The surface V is harmonic between singularities ($\Delta V = 0$), confirming the electrostatic analogy. The zeros are point charges; the critical line is the equilibrium of the boundary conditions imposed by the functional equation's symmetry $s \leftrightarrow 1 - s$.

3.5. **Cross-Domain Validation.** The engine runs on elliptic curve L-functions with zero code changes. For four toy curves ($y^2 = x^3 + ax + b$), the potential V at $\sigma = 1$ discriminates rank: the curve with a rational point (suggesting rank ≥ 1) has $V = -1.32$, deeper than rank-0 curves ($V \approx -1.00$ to -1.23). Canal A (a_p) is orthogonal to all geometric features ($\rho < 0.11$). The engine also maps to Cayley graph spectra: eigenvalues of the graph Laplacian correspond to zeta zeros, the spectral gap corresponds to the zero-free region, and the Ramanujan bound corresponds to RH.

4. Connection to Recent Work

4.1. **Connes (arXiv:2602.04022, February 2026).** Connes [1] extremizes the Weil quadratic form restricted to primes ≤ 13, approximating the first 50 zeros with precision ranging from 2.6×10^{-55} to 10^{-3}. His prolate operator is the formalization of our Module 4: it constructs optimal weights by extremizing a quadratic form. Our decomposition $\theta^* = f(b) \cdot \cos(\omega)$ is the numerical separation of his operator into infrastructure ($f(b)$, amplitude) and signal ($\cos(\omega)$, phase). Our engine achieves 10^{-6} precision with L-BFGS on the log-derivative loss; the 49 orders of magnitude difference reflects the difference between the Weil form and a naive loss function.

4.2. **Conrad (*Canadian J. Math.*, 2005).** Conrad proved that convergence of partial Euler products on the critical line is equivalent to a condition strictly deeper than the Riemann hypothesis. Our computational verification (50-digit Cauchy test, three independent methods) confirms his theorem. Our geometric contribution is making the non-convergence visible as a spiral that never closes.

4.3. **Guth–Maynard (2024, accepted *Annals* 2025).** Their zero density estimate [3] $N(\sigma, T) \leq T^{30(1-\sigma)/13+o(1)}$ improves Ingham's 1940 bound. The result has no direct intersection with our work: their techniques (large value estimates, decoupling) operate in a different space from our geometric features. We include it for completeness, not for borrowed relevance.

5. Attribution and Scope

(a) The non-convergence of partial Euler products at $\sigma = 1/2$ is Conrad [2], formalizing earlier work. Our contribution: computational verification and geometric visualization.
(b) The toroidal embedding via p^{-s} on the Bohr torus is classical (Bohr 1920s). Our contribution: the pair-weighted mixed-torus parametrization and three-scale decomposition.
(c) The identity $f(b) = b/(b+1)$ as the Poincaré compactification in 1D is elementary. Our contribution: identifying it as the per-step contraction of the Euler product, and proving that its geometric application fails (gap grows, cone does not close).

(d) The class number identity $S - k/2 = -h(-p)/2$ is derivable from Dirichlet 1839. Our contribution: the route of discovery (Substrate Scanning through 5 phases) and explicit verification across 961 primes.
(e) The electrostatic interpretation of $\log|-\zeta'/\zeta|$ is implicit in the theory of meromorphic functions. Our contribution: quantitative verification of harmonicity (0.07% median) and the visualization as 3D surface with Laplacian coloring.

No result in this paper is claimed as a new theorem. The contribution is the engine and the methodology that produced it.

6. Methodology: Substrate Scanning

Substrate Scanning is rapid adversarial computational auditing that produces strategic intelligence by systematically eliminating approaches that cannot work. The cycle is: observation $\to$ overclaim $\to$ quantitative verification $\to$ correction $\to$ audited result.

In this paper, the cycle executed 7 times. Each overclaim was produced by the same failure mode: a partial correspondence was generalized before verification of the boundary conditions. Each correction was triggered by a specific quantitative test: doubling N (killed $P(\infty)$), computing reference-independent exponents (killed $\alpha = -0.96$), the dual test (killed the channel bridge), the cone analysis (killed Poincaré convergence).

The methodology's value is not in producing results but in the speed of elimination. Seven approaches that might take months to evaluate analytically were eliminated in a single computational session, each with quantitative evidence. The engine is the distillation of what survived.

7. Applications

7.1. **BSD Verification.** The engine generates 29 geometric features per (σ, t) query for any elliptic curve L-function. A classifier trained on feature vectors near $s = 1$ for curves of known rank could predict rank for curves of unknown rank. The test is binary and falsifiable: does the $V(\sigma)$ profile near $s = 1$ discriminate rank 0 from rank ≥ 1? Toy results (4 curves, $K = 20$) show $V = -1.32$ for the rank-1 candidate vs. $V \approx -1.00$ for rank-0 curves.

7.2. **Optimal Prime Selection.** The decomposition $\theta^* = f(b) \cdot \cos(\ln(p) \cdot \gamma)$ identifies which primes contribute most to detecting a specific zero. For numerical computation of L-function zeros, this reduces cost: 6 well-chosen primes give what 50 generic primes cannot (Connes' result as practical heuristic).

7.3. **Cayley Graph Classification.** The engine maps directly to Cayley graph spectra. Eigenvalues of the graph Laplacian are computed; the spectral gap is measured; the Ramanujan bound is checked. A classifier trained on geometric features could detect expander graphs without full diagonalization—useful in cryptography and network design.

7.4. **Signal Integrity: S-Parameter Classification.** The S-matrix $S(f)$ of a 4-port differential channel is a spectrum parametrized by frequency f. The mixed-mode decomposition into S_{dd} (differential), S_{cc} (common mode), and S_{dc} (mode conversion) maps exactly to Canal B, Canal A, and cross-channel respectively. The motor generates 29 features from $S(f)$ in 0.58 ms per file, classifying channels as PASS/MARGINAL/FAIL at 1709 classifications/second. Validated on 5 synthetic profiles: resonant channels, lossy channels, and high-crosstalk channels are correctly identified with diagnostic reasons. The channel orthogonality test quantifies mode conversion coupling—the signal integrity equivalent of "shared algebra does not imply shared information." For production SI labs measuring 500 boards/day, this replaces manual S-parameter curve inspection with automated classification.

7.5. **Rotating Machinery Fault Diagnosis.** Vibration spectra $V(f)$ from accelerometers on rotating machinery have harmonic orders $n\times$ (analogous to primes), fault frequencies BPFO/BPFI/BSF (analogous to zeros of ζ), and an envelope of harmonic decay (analogous to $f(b)$). The sideband modulation pattern around fault frequencies is $\cos(\omega)$—the phase tuning. The motor generates 29 features from the vibration spectrum at 1132 diagnoses/second (0.88 ms per sample), classifying: normal, unbalance, misalignment, mechanical looseness, bearing outer/inner race defects, and gear mesh faults. Real-time at 50 kHz sampling with 8192-point FFT windows (0.16 s). Severity progression is tracked: a developing bearing defect is detected consistently from SNR = 60 dB through SNR = 20 dB as the number of non-synchronous peaks grows.

7.6. **Physical Cavity Optimization.** The eigenvalues of the Laplacian on a 2D cavity are the resonant frequencies—directly analogous to zeros of ζ. Weyl's law is the analogue of the zero density estimates. The spectral gap determines the quality factor (antenna), the evenness of frequency response (acoustics), or the mode isolation (laser). Tested on 8 cavity shapes (circle, ellipse, rectangle, stadium, parabolic, L-shape): the motor correctly ranks parabolic as optimal for antenna (high Q), stadium for acoustic rooms (uniform spacing), ellipse for laser cavities (large gap), and stadium for quantum billiards (high level repulsion, confirming the Bunimovich stadium as the classical chaotic cavity). Cross-domain comparison shows the ellipse has level spacing ratio $\langle r\rangle = 0.53$, closest to GUE (0.536) and to ζ zeros (0.581)—a numerical echo of the Montgomery–Odlyzko conjecture. Estimated computational savings: factor $\sim$150$\times$ versus full diagonalization when sweeping 10^4 candidate geometries.

7.7. **Colorimetry and Spectral Compression.** A visible spectrum $S(\lambda)$ sampled at 41 points (380–780 nm, 10 nm intervals) is compressed to 3 values by CIE XYZ, losing metamer-distinguishing information. The engine compresses to 29 features preserving spectral structure the eye discards: peak positions, bandwidth, spectral entropy, purity, and an effective metamerism

dimension. Validated on 11 synthetic spectra (pure R/G/B, daylight, incandescent, fluorescent, LED white, ruby, emerald, and two designed metamer pairs). The metamer pair test confirms that feature distance (52.8) exceeds XYZ distance (26.6) by a factor of 2.0×—the features detect differences that XYZ discards. Throughput: 1677 spectra/second (0.6 ms). Standalone engine: `colorimetry_engine.py`.

7.8. **Wavelet 3D Representations and World Models.** Wavelets 3D decompose volumetric data into 8 subbands per level, separating surfaces (curvelets), filaments (ridgelets), and points (isotropic wavelets) at each scale. The motor generates 29 features per subband, producing a tensor of geometric descriptors for the entire volume. Current state of the art (WaveFormer, MICCAI 2025 [4]; WDM, 2024 [5]) uses wavelets 3D as compression layers inside neural networks—the wavelet is a computational tool, not a representation. The gap: no existing work uses wavelets 3D as a deterministic geometric feature generator for world models. The motor fills this gap by producing interpretable, deterministic spectral coordinates from volumetric data before any learning occurs. Applications include: 3D industrial CT (defect classification by topological type—pore vs. crack vs. inclusion), turbulence diagnosis via wavelet scale decomposition (the Kolmogorov cascade is the "Euler product" of the flow), seismic interpretation (faults, channels, and domes as geometric primitives), and 4D flow MRI (vortex detection as "zeros" of the vorticity field). The motor's constraint—the system must have a linear operator with discrete spectrum—is satisfied by wavelets, which linearize multi-scale structure by construction.

7.9. **General Graph Analysis.** The Laplacian $L = D - A$ of any graph has discrete spectrum. The engine computes 29 features including algebraic connectivity (λ_1), spectral gap, level repulsion, partition balance via the Fiedler vector, effective resistance, and hub dominance. Validated on 9 graph configurations: Erdős–Rényi sparse and dense (correctly classified as EXPANDER at high density), Barabási–Albert (SCALE-FREE, FRAGILE), Watts–Strogatz (REGULAR), stochastic block model (MULTI-COMMUNITY), star, path, grid, and a synthetic bot farm. Bot farm vs. organic network separates on 12 features with diff > 0.3—the strongest being health score (1.36 separation), max eigengap position (1.11), and small-world indicator (0.77). Critical node detection via Fiedler vector identifies supply chain vulnerabilities and network bottlenecks. Throughput: 833 graphs/second at $N = 100$, 80 ms at $N = 500$.

7.10. **Abstract Syntax Tree Analysis.** A program's AST is a tree, a tree is a graph, and the engine applies directly. Python source code is parsed to AST via the standard `ast` module, converted to a labeled graph (nodes = operations, edges = parent-child), and the 29 graph spectral features are extracted. 15 code-specific features are added: depth, branching, type entropy, control flow ratio, and function ratio, yielding 44 features per source

file. Validated on 7 code samples ranging from clean functions to deeply nested, spaghetti, obfuscated, and well-structured code. Clone detection correctly identifies fibonacci/clone_fibonacci as structurally identical (similarity = 1.000). Classification detects: DEEP-NESTING (depth = 14), MONOLITHIC (0 functions in 122 nodes), and WELL-MODULAR patterns. Critical AST nodes identify refactoring targets. Current limitation: the raw spectral similarity is too coarse for tree discrimination—all trees have similar Laplacian shape. Normalization by size and relative eigenvalue ratios (analogous to $\theta^*/f(b)$) is the known fix. Throughput: 139 analyses/second.

7.11. **Materials Design via Phonon Spectra.** The dynamical matrix $D(q)$ of a crystal has eigenvalues $\omega^2(q)$—the phonon frequencies. The engine generates 29 features from the phonon spectrum of a 1D chain model: band gap, group velocity, Debye temperature proxy, thermal conductivity proxy, optical branch flatness (rattler modes), and level spacing statistics. Validated on 12 materials ranging from monatomic metals (Cu, Fe, Al) to diatomic semiconductors (NaCl, GaAs, SiC, PbTe), polymers, perovskites, and clathrates (skutterudite proxy). The engine correctly classifies SiC as SEMICONDUCTOR + STIFF (high gap, high v_{sound}), PbTe and BiTe as thermoelectric candidates (low κ, flat optical branches), and polymer-like chains as INSULATOR (large gap from bond alternation). Thermoelectric screening ranks materials by a ZT proxy = $(\text{gap} + 0.1) \times (\text{flatness} + 0.1)/(\kappa + 0.1)$, correctly placing clathrates and PbTe-like materials at the top. Inverse design: 29,241 compositions swept in 32 seconds (905 /s), finding the optimal mass/spring combination for a target spectrum. The cross-domain mapping closes: wavevector q is t, phonon frequencies are zeros, the band gap is the zero-free region, and the Bloch phase e^{-iqa} is $\cos(\ln(p) \cdot \gamma)$.

7.12. **3D Wavelet Subband Analysis.** A 3D discrete wavelet transform (Haar, manually implemented) decomposes volumetric data into 8 subbands per level across 3 levels. The engine generates 29 features from subband energies, kurtosis, entropy, sparsity, anisotropy per level, directional energy decomposition (edge/cross/corner), scale self-similarity, and spectral centroid. Validated on 6 synthetic volumes: sphere, cylinder, two spheres (multi-scale), Gaussian noise, sphere+noise (denoising), and torus. The classifier correctly separates STRUCTURED from NOISY (kurtosis + sparsity) and identifies MULTI_SCALE objects. Cylinder and torus are ambiguously classified as MULTI_SCALE because they possess two inherent length scales; this ambiguity is genuine, not a failure. Throughput: 50 volumes/second at 32^3. The cross-domain mapping: subbands are primes, subband energy is $p^{-\sigma}$, energy decay across levels is $f(b)$, and anisotropy per level is $\cos(\ln(p) \cdot \gamma)$.

7.13. **Turbulence Cascade Classification.** Multi-level 1D Haar DWT decomposes velocity signals into per-scale energy and kurtosis profiles. The engine generates 29 features: cascade energy ratios, K41 deviation (theoretical $2^{5/3}$ ratio), kurtosis growth rate across scales, spectral flatness, cross-scale

magnitude correlation, and self-similarity. Validated on 5 synthetic regimes: laminar (sinusoidal), Kolmogorov ($k^{-5/3}$ filtered noise), intermittent (Kolmogorov + 5× bursts), white noise (broadband), and tonal (machine-like harmonics). The classifier achieves 5/5 correct using kurtosis sign (sub-Gaussian for sinusoids), burst detection (extreme kurtosis for intermittency), and spectral slope (negative for broadband). Limitation: Haar has poor frequency resolution; the K41 cascade ratio is not recovered cleanly. Real turbulence analysis requires Daubechies or Morlet wavelets. Throughput: 476 signals/second at 4096 samples. The cross-domain mapping: wavelet scales are primes, $E(j)$ is $p^{-\sigma}$, cascade ratio is $f(b)$, and kurtosis per scale is $\cos(\ln(p) \cdot \gamma)$.

7.14. **Image Spectral Analysis.** 2D FFT with azimuthal averaging yields a radial power spectrum $P(f)$ from grayscale images. The engine generates 29 features: spectral slope (natural images follow $1/f^2$, per Field 1987), band energies, peak detection, angular isotropy, spectral centroid, and forgery anomaly scoring. A forgery detection module compares local vs. global spectral slope across 16×16 blocks to identify pasted regions. Validated on 6 synthetic images: white noise (slope ≈ 0), natural-like (slope ≈ -2), periodic texture, step edge, noisy natural, and forged (natural + pasted periodic block). Classifier: 6/6 correct. The forged image is correctly identified as TEXTURED at the global level because the pasted periodic block injects spectral peaks. Limitation: 64×64 pixels; real forensics needs megapixel images with multi-scale block analysis. Throughput: 816 images/second. The cross-domain mapping: spatial frequency is t, spectral peaks are zeros, $1/f^2$ decay is $f(b)$, and angular sector distribution is $\cos(\ln(p) \cdot \gamma)$.

7.15. **Spectral Pathfinding.** Building on `graph_engine.py` (§7.9), this engine adds pathfinding-specific features: spectral coordinates from Laplacian eigenvectors, effective resistance via pseudoinverse, BFS vs. spectral distance rank correlation, and algorithm recommendation. The engine generates 29 features. Validated on 5 graph types: 10×10 grid (A_STAR_SPECTRAL, rank correlation 0.71), Erdős–Rényi 100-node (BFS, well-connected), bottleneck SBM (BRIDGE_FIRST, high eigengap), Barabási–Albert (ROUTE_HUBS, scale-free), and 50-node path (A_STAR_SPECTRAL). The spectral heuristic is most valuable for graphs with bottleneck structure where eigenvector embedding reveals global geometry that BFS cannot see. For regular/expander graphs, BFS is already optimal. Throughput: 22 graphs/second at $N = 100$. The cross-domain mapping: node index is t, critical nodes (Fiedler) are zeros, spectral coordinates are the torus embedding, and effective resistance is $f(b)$.

7.16. **Elliptic Curve Cryptographic Auditing.** For elliptic curves $y^2 = x^3 + ax + b$ over $\mathbb{F}_p$, brute-force point counting gives $\#E(\mathbb{F}_p)$ and $a_p = p + 1 - \#E$. The engine generates 29 features from group order factorization (Canal A: largest prime factor ratio, smoothness, cofactor), embedding

degree (Cross: MOV/Frey–Rück vulnerability), Frobenius eigenvalue structure, anomalous flag ($\#E = p$), and a composite security score. Validated on 5 curves: secure ($y^2 = x^3+3x+2$ over $\mathbb{F}_{127}$, prime order 139, $\text{emb}_k = 69$), Pohlig–Hellman (16-smooth order), anomalous ($\#E = p$), MOV-weak (supersingular, $\text{emb}_k = 2$), and supersingular ($a_p = 0$). Classifier: 5/5 correct, with anomalous/MOV/supersingular correctly flagged as WEAK with multiple issues. Limitation: brute-force counting is $O(p^2)$, limiting p to $\sim$500. Real cryptographic curves use $p > 2^{200}$. The cross-domain mapping: group order is the spectral gap, factorization structure is Canal A, embedding degree is the Ramanujan bound, and $a_p/(2\sqrt{p})$ is $\cos(\ln(p) \cdot \gamma)$.

8. Open Questions

Note: §§7.9–7.11 were validated after the initial draft, confirming zero-change transfer.

(a) Does the R^2 of the factored form $\theta^* = f(b) \cdot \cos(\omega)$ improve with the Weil quadratic form as loss instead of the log-derivative? If $R^2 \to 1$, the decomposition is exact and $f(b) \cdot \cos(\omega)$ is the prolate eigenvector in closed form.

(b) Is the third axis ($\cos(\ln(p) \cdot \gamma)$) sufficient, or does a fourth axis emerge at higher K? The residual after $A+B+C$ projection is 55% of norm at $K = 30$. A neural network with more capacity might reduce this, or it might be irreducible noise.

(c) Does the $V(\sigma)$ profile at $s = 1$ discriminate elliptic curve rank at scale? The toy (4 curves, $K = 20$) is suggestive but far from conclusive. LMFDB provides thousands of curves with certified rank.

(d) Can the harmonicity of $V(\sigma, t)$ be improved by using the Weil form instead of the log-derivative? The 0.07% residual may be inherent to the partial product or may be reducible with better operators.

9. Deliverables

(a) `geometric_engine.py` — self-contained Python engine (numpy/scipy only), 8 modules, 29 features, $\sim$8 ms per query. Supports $\zeta(s)$, $L(s, \chi)$, $L(s, E)$ with `EllipticCurveL` class. AGPL-3.0.

(b) `sparam_engine.py` — S-parameter spectral engine (1709/s).

(c) `motor_diagnosis.py` — vibration fault diagnosis (1132/s).

(d) `graph_engine.py` — general graph spectral analysis (833/s at $N = 100$).

(e) `ast_engine.py` — Python AST spectral analysis, $29 + 15 = 44$ features (139/s).

(f) `cavity_engine.py` — physical cavity optimization, 8 shapes, application scoring (20/s).

(g) `phonon_engine.py` — 1D chain phonon spectra, 12 materials, thermoelectric screening (478/s).

(h) `colorimetry_engine.py` — visible spectrum analysis, metamer detection, 2× XYZ discrimination (1677/s).
(i) `wavelet_3d_engine.py` — 3D Haar DWT, 6 synthetic volumes at 32^3, structural classification (50/s).
(j) `turbulence_engine.py` — 1D Haar DWT, 5 turbulence regimes, cascade analysis (476/s).
(k) `image_spectral_engine.py` — 2D FFT radial spectrum, forgery detection, 6 image types (816/s).
(l) `pathfinding_engine.py` — spectral graph pathfinding, algorithm recommendation, builds on `graph_engine` (22/s at $N = 100$).
(m) `crypto_curve_engine.py` — elliptic curve security audit over $\mathbb{F}_p$, 5 vulnerability classes (274/s at $p = 101$).

All engines share the same 29-feature core architecture.

Additional deliverables:

(xiv) `envelope_pipeline.py` (v1) and `envelope_v2.py` — optimization pipelines for envelope weights. L-BFGS and neural network variants.
(xv) 17 interactive HTML visualizations (Three.js r128): toroidal embeddings, torus modes, operators, landscapes, spectra.
(xvi) `dual_role_f_v3.docx` — technical report on the three scales, $f(b)$, and Poincaré compactification, with all 7 overclaims documented and corrected.
(xvii) `class_number_report.docx` — the identity $S - k/2 = -h(-p)/2$, route of discovery, and 961-prime verification.

10. Conclusion

The engine is the product; the results are the byproducts. No individual result survived as mathematical novelty against the existing literature. What survived is a methodology that eliminates impossible approaches at visual speed, a decomposition that separates infrastructure from signal in any spectrum, and a modular tool that generates geometric parameters without statistical assumptions. The architecture transfers with zero code changes from $\zeta(s)$ to elliptic curves, Cayley graphs, differential S-parameters, vibration spectra, physical cavities, and visible-light spectroscopy—because the spectrum is the spectrum.

The central finding, confirmed by three independent tests across two L-function families, is that the arithmetic structure of $p - 1$ is orthogonal to the spectral structure of the Euler product phases. Shared algebra does not imply shared information. The operator $f(b) = b/(b + 1)$ acts identically in both channels but transports information in only one. This orthogonality constrains which mathematical structures can and cannot contribute to the detection of zeta zeros, and may inform the design of future operators.

The engine does not prove theorems, replace domain physics, predict temporal evolution, or handle strongly nonlinear systems without wavelet linearization. It generates features, not demonstrations. Its value is in filtering:

generating cheap geometric coordinates that concentrate expensive computation on the few candidates that matter.

The phrase that closes the arc: t parametrises the critical line. Everything else is scaffolding.

$$p^{-s} = p^{-\sigma} \cdot e^{-i\ln(p)\cdot t}$$

This equation is a fact, not a choice. $p^{-\sigma}$ is the amplitude. $e^{-i\omega t}$ is the phase. The zeros are the values of t where all phases conspire. $f(b)$ organises the amplitude. $\cos(\omega)$ organises the phase. Canal A is orthogonal to both. Everything in this paper—the embedding, the three scales, the potential, the 16 domains, the motor—is scaffolding around this single factorisation.

A note on democratisation. The engine lowers the barrier of access: anyone with Python and a browser can explore ζ, diagnose a turbine, or screen a material. What it does not lower is the barrier of judgement. The 7 overclaims in §0 were produced by the same tool that produced the surviving results. Detecting them required domain knowledge the tool does not have. Accessible tools without accessible scepticism produce false confidence. The real democratisation in this paper is not the motor—it is §0. The errata-first protocol teaches that the tool fails, when it fails, and how to detect the failure. *Democratising scepticism is worth more than democratising computation.*

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